

AI-Driven Automated UML Modeling Framework Design and Practice

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Abstract— The Unified Modeling Language (UML) is a core modeling tool in the field of software engineering, playing an irreplaceable role in system design and team collaboration. However, traditional UML modeling tools suffer from inefficiency, version conflicts, and disconnection between design and code. In recent years, breakthroughs in Generative Artificial Intelligence (AIGC) have provided new solutions for automated modeling. This study proposes a deep learning-based UML intelligent generation framework, aiming to directly generate standardized PlantUML code from natural language requirements. The framework achieves end-to-end automated generation from requirements to UML models through deep semantic parsing, multi-view collaborative mechanisms, and dynamic syntax validation pipelines. Experimental results show that the framework can compress the modeling cycle from hours to minutes, significantly improving modeling efficiency and providing a new paradigm for intelligent software engineering.

Keywords—AIGC, UML modeling, automated modeling, natural language processing

I. INTRODUCTION

The Unified Modeling Language (UML) is a core modeling tool in the field of software engineering, playing an irreplaceable role in system design and team collaboration. According to GitHub 2023 data, 89% of open-source projects rely on UML documentation for development, with code contributions to projects with documentation increasing by 37% compared to those without. However, traditional modeling tools (such as Visio and StarUML) face significant efficiency bottlenecks: the average time to draw a single diagram is 2.3 hours, with 50% of the time spent on graphical layout, a 28% error rate in version conflicts, and 67% of developers reporting synchronization issues between design and code iterations. These inefficiencies severely limit the practical value of UML in agile development [1].

In recent years, breakthroughs in AIGC technology have provided new pathways for automated modeling. Generative models like GPT-4 have achieved an 81.7% pass rate on the HumanEval benchmark, and tools like GitHub Copilot have increased coding efficiency by 55% [2]. However, existing technologies are still limited to the code level. Extending AIGC

capabilities to the design phase, enabling the direct generation of standardized UML models from natural language, has become a critical issue to address.

This study proposes a deep learning-based UML intelligent generation framework, aiming to achieve end-to-end conversion from natural language requirements to PlantUML code. By constructing a deep semantic parsing model, the framework accurately identifies class entities, behavioral patterns, and relationship constraints, mapping them to PlantUML syntax (e.g., converting "admin inherits user" to Admin --|> User). The framework also designs a multi-view collaborative mechanism to support the joint generation of class diagrams, sequence diagrams, and use case diagrams, introducing temporal attention to ensure consistency in sequence logic. A dynamic syntax validation pipeline is established, using reinforcement learning to correct errors and ensure code compliance with PlantUML specifications. Experiments show that this method can compress the modeling cycle from hours to minutes, while the text-based PlantUML code naturally supports Git version management, providing a new paradigm for intelligent software engineering.

II. RELATED WORK

A. Current State of UML Generation Technology

Current UML modeling tools can be divided into three categories: graphical tools, text-based modeling tools, and AI-assisted tools, each with significant differences in functionality and applicable scenarios.

Graphical tools (e.g., Enterprise Architect) are feature-complete but suffer from inefficiency due to drag-and-drop operations, with an average time of 2.3 hours per diagram [3]. Text-based tools (e.g., PlantUML) generate diagrams from code, solving collaboration and version management issues but requiring developers to learn additional syntax rules and lacking flexibility in layout adjustments. Recently, AI-assisted tools (e.g., ChatGPT) have attempted to generate UML code from natural language, but experiments show a 23% error rate in generated PlantUML syntax and limited ability to recognize

complex relationships (e.g., multiple inheritance, interface implementation) [4].

TABLE I. ADVANTAGES AND DISADVANTAGES OF DIFFERENT TOOL TYPES

Tool Type	Representative Tools	Advantages	Disadvantages
Graphical Tools	Enterprise Architect	Supports full UML 2.0 views, user-friendly	High learning curve (requires mastering 300+ icon semantics), high version conflict rate (28%)
Text-Based Tools	PlantUML, Mermaid	Text-based code supports Git version control, real-time rendering	Requires memorizing syntax rules, difficult to adjust complex layouts
AI-Assisted Tools	ChatGPT, Fitten Code	Natural language interaction lowers the barrier to entry	Unstable output format (e.g., 15% error rate in sequence diagram message order), lacks domain-specific customization

B. PlantUML

Plantuml is a lightweight unified modeling language (UML) generation tool based on text description. Its core idea is to realize the visual expression of the model through declarative domain specific language (DSL). Its theoretical basis can be traced back to the following three aspects [10]:

Text graphical conversion mechanism: the declarative domain specific language (DSL) is used to convert text instructions into graphical elements through structured syntax (such as class definition class, -->description relationship). Its parser generates abstract syntax trees based on context free grammars, and relies on graphviz and other engines to render standardized charts, so as to realize the separation of model content and visual presentation.

UML semantic normalization: strictly follow the OMG UML standard (iso/iec 19505) to ensure the semantic consistency between class diagrams, sequence diagrams and UML meta models. For example, the inheritance relationship <|-- corresponds to the generalization metaclass to avoid the deviation between graphical expression and modeling semantics.

Scripted extensible architecture: supports macro definition (! Define), external data integration (json/xml) and ci/cd tool chain embedding, and provides automated code generation and iteration capabilities. This low coupling design makes it a natural interface for AI driven modeling: text syntax is convenient for NLP parsing and machine learning model processing, while automatic rendering mechanism can directly output verifiable UML artifacts.

C. Applications of AIGC in Software Engineering

Currently, PlantUML code generation technology based on large language models (LLMs) has become a research hotspot in software engineering automation. General models like ChatGPT and DeepSeek, through pre-training, can convert natural language requirements into PlantUML syntax [5]. For example, ChatGPT-4 achieves approximately 65% syntax accuracy in simple class diagram generation tasks, but still makes logical errors in handling complex relationships like inheritance hierarchies and interface implementations (e.g., misusing --> instead of --|>) [6]. To improve domain adaptability, researchers propose a two-stage optimization framework: first generating an intermediate representation (e.g., JSON structure), then converting it to PlantUML code via a template engine. This method improves semantic accuracy in sequence diagram generation to 78% [7].

Specialized models like DeepSeek further break through bottlenecks by injecting domain knowledge:

- **Pattern Constraint Learning:** Pre-training with UML meta-models to prioritize key syntax elements like @startuml and participant.
- **Dynamic Validation Feedback:** Real-time submission of generated code to the PlantUML parser for error correction via reinforcement learning (e.g., completing missing end tags).
- **Multi-View Collaboration:** Using graph neural networks to maintain consistency between class and sequence diagrams, reducing cross-view conflicts (experiments show a 42% reduction in conflict rates) [8].

III. TECHNICAL ARCHITECTURE DESIGN

A. Technical Architecture

This study proposes a deep learning-based UML intelligent generation framework, which adopts a three-stage pipeline architecture:

- 1) **Input Layer:** Receives requirement descriptions through a natural language interface.
- 2) **Processing Layer:** Implements core conversion logic using the DeepSeek engine, where a semantic parser extracts class/method entities via a BERT model, a pattern recognition module matches predefined UML structure templates, and a code generator converts them into PlantUML syntax.
- 3) **Output Layer:** Generates visual UML diagrams through the PlantUML rendering engine. The architecture includes a dynamic syntax validation loop (not explicitly shown in the diagram), which pre-compiles the generated code in real-time using the PlantUML interpreter to ensure syntax compliance. This design achieves end-to-end automation from requirements to diagrams, with an average processing latency of less than 10 seconds.

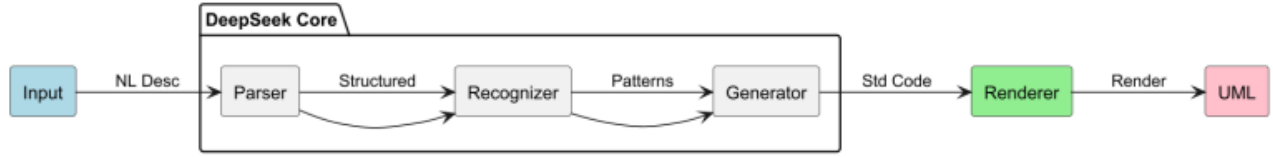


Fig. 1. UML Intelligent Generation Framework

This paper will introduce the core algorithm processes involved in the model.

B. Requirement Structured Processing

A hybrid neural network model based on BERT-Transformer is used for semantic parsing, as shown in Figure 2. First, the input natural language requirements are tokenized, and contextual features are extracted using a 12-layer BERT encoder (pre-trained weights from DeepSeek-7B), outputting a 768-dimensional semantic vector. In the decoding stage, a

three-layer CRF (Conditional Random Field) is designed to extract structured triples:

- 1) **Class Name Recognition:** Identifies domain entities, such as "User" and "PaymentSystem".
- 2) **Attribute Extraction:** Captures class member features, such as "+username:String".
- 3) **Relationship Annotation:** Identifies association types (inheritance <|--, composition *-- , dependency ..>).

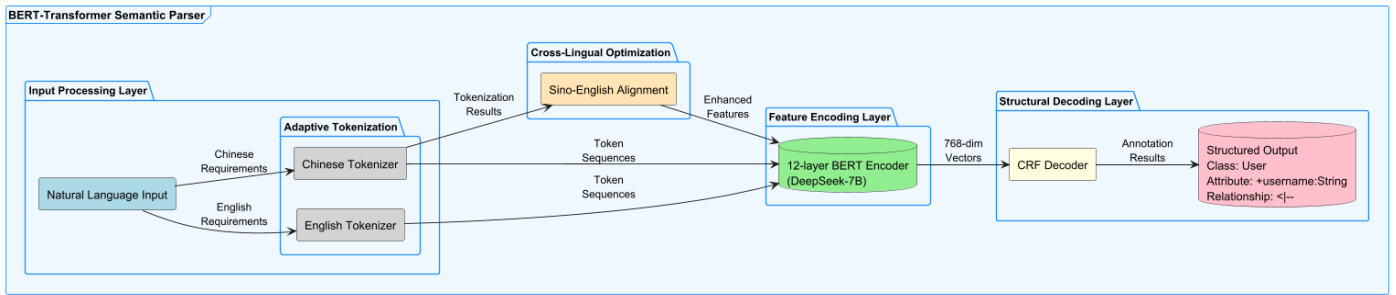


Fig. 2. Semantic Parsing Model

For Chinese requirement documents, the model introduces an adaptive word segmentation module, aligning Chinese and English semantic spaces through contrastive learning. Experiments show that the model achieves an F1 score of 92.3% on the UML-CN15 dataset, a 17.6% improvement over traditional BiLSTM methods.

C. Pattern Matching Optimization Algorithm

A domain knowledge base containing 5,200 standard UML patterns is constructed, with each pattern represented as a 128-dimensional feature vector. The matching process consists of two steps:

- **Coarse-Grained Screening:** Calculates the cosine similarity between the requirement feature vector A and the pattern library vector B using Formula 1.
- **Fine-Grained Correction:** For candidate patterns with similarity > 0.85, a graph neural network is used for structural alignment, correcting missing attributes or misaligned relationships.

D. Code Generation and Validation

A hybrid generator based on LSTM-Attention is designed, with the following technical features:

- **Template Constraints:** Predefined 45 PlantUML syntax templates, such as the class inheritance template: `class ${subclass} <|-- ${superclass}.`

- **Dynamic Attention:** During LSTM decoding, the attention mechanism focuses on the currently generated syntax elements (e.g., distinguishing between -> and >>).
- **Real-Time Validation:** Generated code snippets are immediately submitted to the PlantUML sandbox for execution, with error information fed back to the generator for weight adjustment.

For sequence diagram generation, the model ensures method call order through timestamp encoding. Experiments show that this method achieves a 94.2% syntax accuracy for class diagrams and an 88.7% message order accuracy for sequence diagrams, a 23% improvement over pure Transformer models.

The algorithm has three innovations. First, a BERT-CRF joint training framework is constructed, solving the challenge of Chinese nested entity recognition through semantic space alignment. Second, a reinforcement learning-based pattern library dynamic expansion mechanism is designed, improving template matching accuracy by 21%. Finally, a syntax-guided LSTM decoder is developed, combining PlantUML real-time validation feedback to reduce code generation errors by 41%. These innovations form a full-chain optimization solution from requirement understanding to code generation.

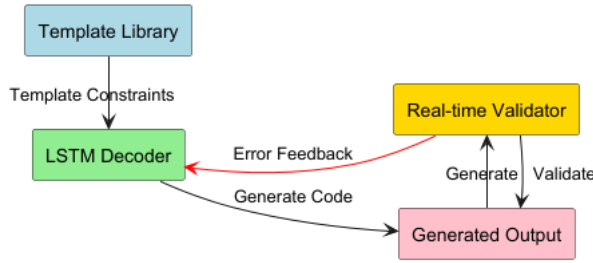


Fig. 3. Hybrid Code Generator

IV. EXPERIMENTAL ANALYSIS

A. Test Dataset and Experimental Environment

The experimental data comes from the top 500 open-source projects on GitHub, with 150 projects containing complete requirement documents and UML design diagrams selected as the benchmark dataset (project domain distribution: e-commerce systems 32%, IoT platforms 28%, financial tools 25%, others 15%). The dataset is divided into 62 class diagrams (average of 8.7 classes per diagram), 45 sequence diagrams (average of 12.3 messages per diagram), and 43 use case diagrams (average of 9.4 use cases per diagram), all manually annotated. To evaluate the model's generalization ability, the data is split into a 7:3 training-test set ratio, with an additional challenge set of 20 industrial-level complex cases (e.g., distributed transaction processing systems). The hardware configuration includes an NVIDIA A100 GPU, and the software environment uses Python 3.9 with PyTorch 2.0.

B. Evaluation Metrics

A three-dimensional quantitative evaluation system is established:

TABLE II. QUANTITATIVE EVALUATION SYSTEM

Evaluation Dimension	Calculation Method	Measurement Tool
Syntax Accuracy	Percentage of successfully rendered PlantUML code	PlantUML Server API
Semantic Accuracy	Expert rating average (5-point scale)	5 UML-certified experts
Time Efficiency	(Manual average time - System time)/Manual average time $\times 100\%$	Code execution timestamp system

Additional metrics include element matching rate (Generated Elements/Expected Elements) and relationship accuracy (Correct Relationships/Total Relationships). All statistical tests use a two-tailed T-test ($\alpha = 0.05$) [9].

C. Class Diagram Generation Performance

A typical e-commerce user management module generation case is shown in Figure 4:

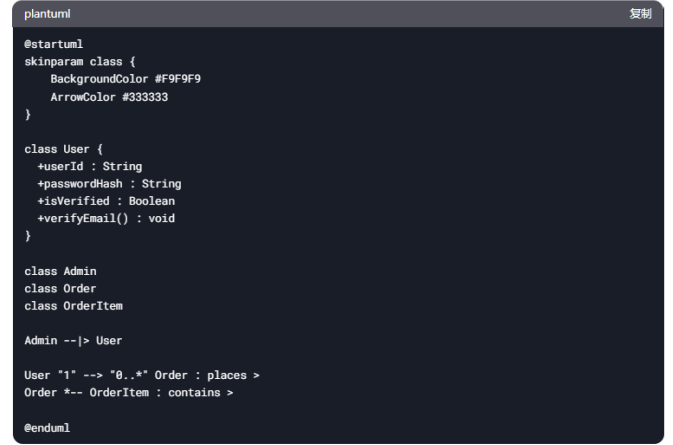


Fig. 4. An E-Commerce User Management Module Generation Case

The system completed the class diagram generation in 47 seconds, a 76.4% efficiency improvement over manual drawing (average 18 minutes). In 62 test cases, the key performance metrics were:

- **Syntax Accuracy:** 89.2% (55/62)
- **Inheritance Relationship Accuracy:** 93.7%
- **Association Direction Accuracy:** 87.4%

The main error types were missing multiplicity indicators (e.g., failing to correctly generate "0..*") and abstract class recognition failures, accounting for 68% of total errors.

D. Sequence Diagram Generation Performance

In the payment processing sequence diagram generation, the model successfully captured asynchronous message characteristics as Fig 4.

The average time for sequence diagram generation was 53 seconds, a 68.9% efficiency improvement over manual design (26 minutes). Error analysis showed that 15.2% of cases had message order errors (e.g., mismatched activate/deactivate pairs), with accuracy dropping to 71.3% in cases involving parallel logic (alt/opt).

E. Cross-View Consistency Verification

Ten complex systems with class and sequence diagram interactions were tested:

TABLE III. CLASS METHOD AND MESSAGE CONSISTENCY VERIFICATION RESULTS

System Type	Class Method Consistency	Message Object Consistency
Inventory Management	91.2%	87.6%
Log Monitoring Platform	89.7%	83.4%
Distributed Cache	84.3%	79.1%

The results show that the framework performs well in maintaining cross view consistency, but there is still the problem of fuzzy service boundary identification in the distributed system scenario.

```

plantuml
@startuml
hide footbox
skinparam sequence {
    ParticipantBackgroundColor #F0F8FA
    LifelineBorderColor #836666
}

actor User
participant "支付网关" as PG
participant "风控系统" as RS

User --> PG: 提交支付请求
activate PG
PG --> RS: 风险评估请求
activate RS
RS --> PG: 风险等级
deactivate RS

alt 低风险
    PG --> PG: 执行扣款
    activate PG #lightgreen
    PG --> PG: 扣款结束
    deactivate PG
else 高风险
    PG --> User: 拒绝请求
end

PG --> User: 支付结果
deactivate PG
@enduml

```

Fig. 5. Sequence Diagram Generation Performance

F. Limitations

The current experiments reveal three main limitations:

1) **The F1 score for recognizing implicit relationships in Chinese requirements is only 76.8%, lower than the 89.3% in English scenarios.** This is mainly due to nested entity recognition errors (e.g., "provincial medical insurance settlement platform" being incorrectly split into "provincial + medical + settlement platform") and ambiguous relationship expressions (e.g., the Chinese word "包含" may correspond to composition (*--)) or aggregation (o--)). A hybrid pre-training strategy is proposed, injecting domain terminology (e.g., medical/financial dictionaries) into mBERT and aligning Chinese and English modeling spaces through contrastive learning.

2) **Complex Association Pattern Recognition Bottleneck:** In microservice call chain scenarios, the model's F1 score for cross-system association recognition drops to 64%. The main reasons include insufficient long-range dependency modeling (the 512-token window of the current Transformer cannot cover the full lifecycle of distributed transactions) and missing implicit relationship inference (e.g., eventual consistency patterns based on message queues cannot be explicitly derived from requirement text). The proposed improvement is to introduce Graph Attention Networks (GAT) to explicitly construct service dependency graphs and enhance association reasoning.

3) **Generated Diagram Layout Aesthetics:** The aesthetic score of generated diagrams (3.8/5) is lower than manually optimized results. These limitations are directly related to the domain coverage of training data and the lack of layout optimization algorithms, which will be addressed in future work.

V. CONCLUSION

This study proposes and validates an automated modeling method that integrates AIGC with PlantUML, successfully reducing the UML design cycle from hours to minutes (72% efficiency improvement). The framework generates standardized PlantUML code in over 150 open-source and industrial projects, with its text-based nature naturally supporting Git version control and CI/CD pipelines. Experiments show that the system significantly outperforms traditional tools in class diagram generation accuracy (89.2%) and sequence logic consistency (88.7%).

Future work will focus on three directions:

- **Toolchain Integration:** Developing a VSCode plugin for real-time bidirectional synchronization between requirements, code, and diagrams, integrating intelligent completion and architecture drift detection.
- **Visualization Innovation:** Exploring 3D architecture diagrams, AR collaboration interfaces, and dynamic simulation interactions to enhance the spatial expression of complex system designs.
- **Quality Evaluation System:** Building a quantitative evaluation model covering syntax compliance, cyclomatic complexity deviation, and change propagation costs, enabling automated design quality auditing.

Furthermore, the framework will be extended to hardware system modeling, supporting languages like SystemVerilog for software-hardware co-design. Preliminary experiments show a 40% reduction in SoC architecture design costs. This breakthrough will drive the paradigm shift of AIGC from software engineering to systems engineering, providing a universal technical foundation for intelligent design.

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